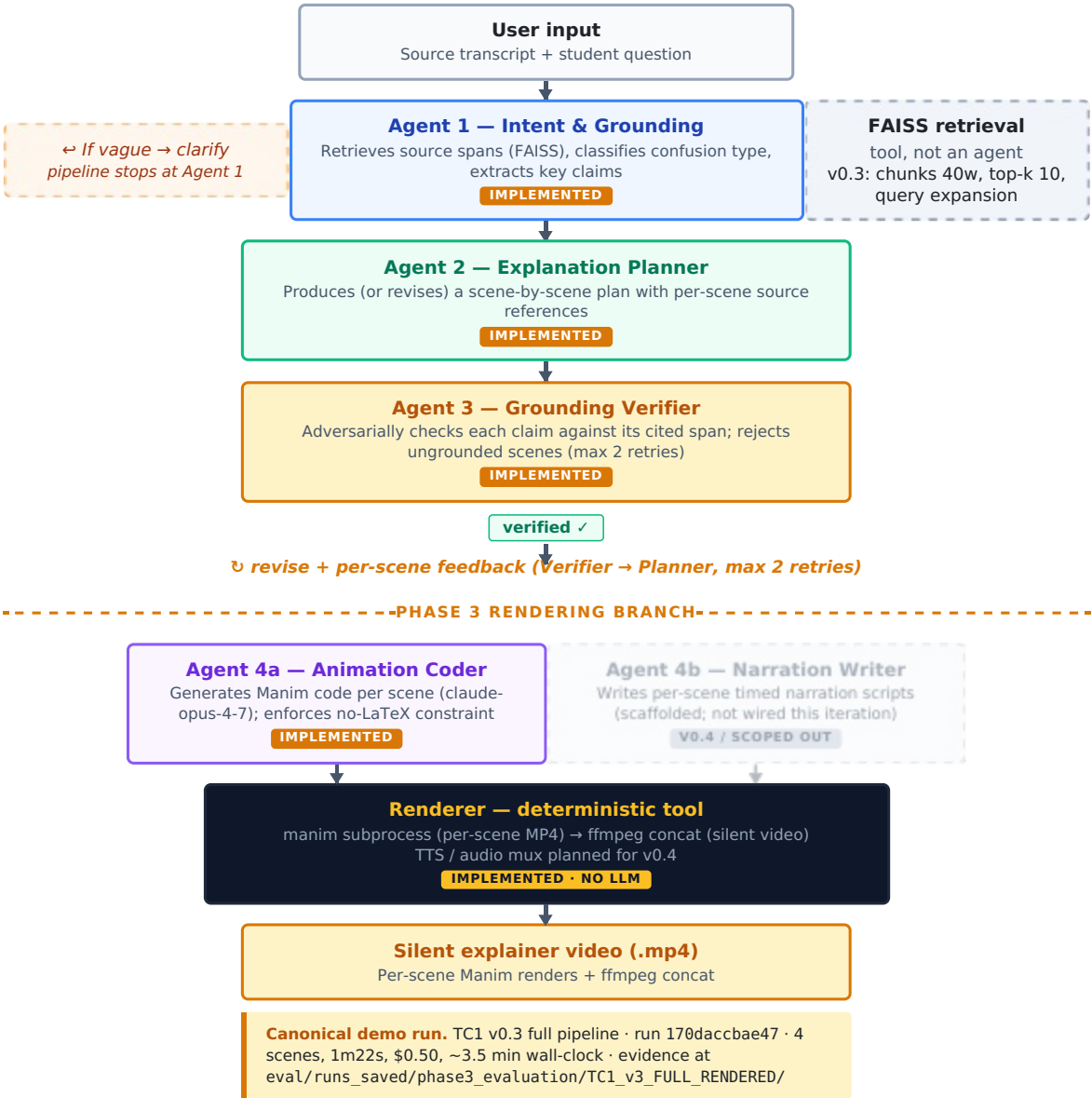


Aurua — Architecture Diagram

PHASE 3 · V0.3 · APRIL 29, 2026

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Legend

Agent 1	Intent & Grounding. Implemented. FAISS retrieval as a tool; classifies confusion type; can short-circuit on vague input. v0.3 adds query expansion.
Agent 2	Explanation Planner. Implemented. Produces structured scene plans; re-plans flagged scenes on verifier feedback (max 2 retries).
Agent 3	Grounding Verifier. Implemented. Distinct system prompt; checks each claim against its cited span. Caught real planner overreach in 3 of 12 evaluation runs.
Agent 4a	Animation Coder. New in Phase 3. One Claude-Opus-4-7 call per scene. Strict no-LaTeX constraint baked into the prompt.
Agent 4b	Narration Writer. Scaffolded but not wired this iteration; reserved for v0.4 (paired with TTS + ffmpeg muxing).
Renderer	Deterministic tool, no LLM. New in Phase 3. manim CLI per scene → ffmpeg concat to single silent MP4.

Coordination Notes

Custom orchestrator (not LangGraph). Control flow lives in <code>src/run.py</code> — ~80 lines of plain Python, linear with one back-edge. LangGraph adds setup cost without saving meaningful code at this topology size. Justified in final report § 2.4.
Verifier feedback loop. The amber back-arrow makes this genuinely agentic rather than a prompt chain. Per-scene verdicts; only flagged scenes revise. After 2 failed retries, exit with <code>verification_exhausted</code> rather than ship ungrounded content (4x across the 12-run set).
Phase 3 branch is conditional. Rendering only runs when verification passes. Cost cap held: no run exceeded 10 API calls or \$0.50, including full-rendering runs.
Shared state via Pydantic. Single <code>RunState</code> typed model, serialized at every status transition. Per-agent typed views planned for v0.4.